

EXPERIMENT 23:

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Clusterer
Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll -cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Nom) class
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)
21:10:32 - SimpleKMeans
21:16:43 - EM

Cluster output

mean	3.1769	3.0662	2.6547	3.7037	2.3766
std. dev.	0.2599	0.2891	0.2687	0.2857	0.2512
petallength					
mean	1.4194	5.7859	4.7484	1.5173	3.863
std. dev.	0.1692	0.4745	0.3193	0.1592	0.3516
petalwidth					
mean	0.1948	2.1327	1.5757	0.3028	1.1696
std. dev.	0.0557	0.2359	0.2196	0.1212	0.1351

Time taken to build model (full training data) : 0.4 seconds

=== Model and evaluation on training set ===

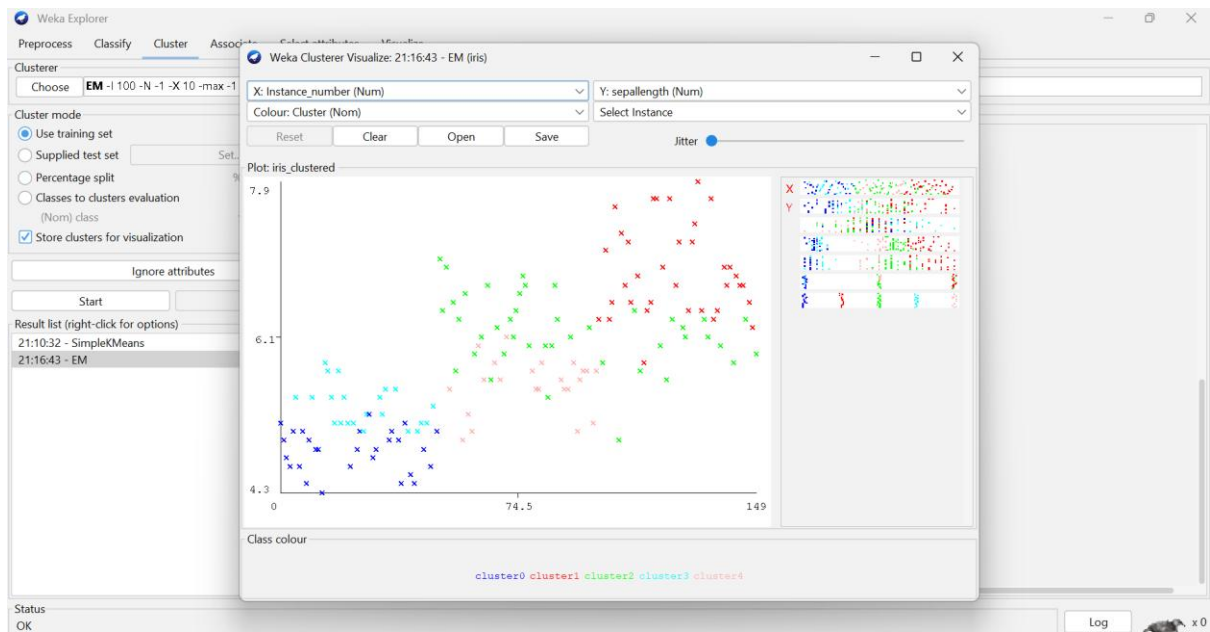
Clustered Instances

0	28 (19%)
1	35 (23%)
2	42 (28%)
3	22 (15%)
4	23 (15%)

Log likelihood: -1.60803

Status
OK

Log



Clusterer output

=== Run information ===

Scheme: weka.clusterers.EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 1
 Relation: iris
 Instances: 150
 Attributes: 5
 sepalength
 sepalwidth
 petallength
 petalwidth
 Ignored: class
 Test mode: evaluate on training data

=== Clustering model (full training set) ===

EM
 ==

Number of clusters selected by cross validation: 5
 Number of iterations performed: 16

	Cluster				
Attribute	0	1	2	3	4
	(0.18)	(0.23)	(0.28)	(0.15)	(0.15)

Clusterer output

std. dev.	0.2405	0.5228	0.4138	0.2407	0.3159
sepalwidth					
mean	3.1789	3.0862	2.8547	3.7037	2.5786
std. dev.	0.2599	0.2891	0.2687	0.2857	0.2512
petallength					
mean	1.4194	5.7859	4.7484	1.5173	3.863
std. dev.	0.1692	0.4745	0.3193	0.1592	0.3516
petalwidth					
mean	0.1948	2.1327	1.5757	0.3028	1.1696
std. dev.	0.0557	0.2359	0.2196	0.1212	0.1351

Time taken to build model (full training data) : 0.4 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	28 (19%)
1	35 (23%)
2	42 (28%)
3	22 (15%)
4	23 (15%)

